

Solution Requirements

Date	7 July 2026
Team ID	SWETID-2026-5556
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional Requirements and Non-Functional Requirements:

The system requirements for the Credit Card Approval Prediction System are split into two core categories to define what the platform does and how it performs.

- Functional Requirements (FR): Define the exact features, functions, and behaviours the system must execute.
- Non-Functional Requirements (NFR): Specify the performance, security, and quality criteria the system must meet.

Step 1: Functional Requirements (FR)

S No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users (Applicants and Bank Admins) must securely log in using email verification and secure password hashing.	High
2	Authorization levels	Two clear roles: Applicant (can submit data and view their own status) and Bank Admin (can view reports and manage model thresholds).	High
3	External interfaces	The system connects via APIs to third-party Credit Bureaus to fetch financial credit scores and histories instantly.	High
4	Transactions processing	The system safely processes application submissions, exchanging encrypted user documents and income profile parameters.	Medium

5	Reporting	Generates an interactive analytics dashboard for admins showing approval rates, rejection reasons, and model prediction metrics.	Medium
6	Business rules, etc.	Applications with incomplete required details (e.g., missing SSN or income data) are blocked before sending to the ML model.	High
7	Compliance to laws or regulations	Adheres to strict data privacy standards like GDPR or financial regulations to securely encrypt sensitive applicant data.	High
8	Other	Includes an automated email and SMS notification system to alert applicants immediately once a final decision is generated.	Low

Step 2: Non-Functional Requirements (NFR)

S No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system must quickly process the applicant's details and return an ML risk score recommendation.	Page loads and ML risk prediction results must render in less than 2 seconds.
2	Scalability	The application architecture must effortlessly handle traffic surges during peak application hours.	Supports up to 10,000 concurrent users checking their application status at the same time.
3	Security & Data Privacy	All sensitive user data, financial profiles, and system databases must be completely protected.	End-to-end AES-256 encryption for all data stored and transmitted through the portal.

4	Reliability & Availability	The online portal must remain up and running without unexpected crashes or system shutdowns.	Guarantees a minimum of 99.9% uptime per month for the applicant portal.
5	Usability & Accessibility	The user interface (UI) must be intuitive, easy to navigate, and fully clean for all applicants.	Fully complies with WCAG 2.1 AA standards for web content accessibility.
6	Other	The application code structure should be easy to maintain, debug, and upgrade with updated ML models.	Built on a modular microservices framework to ensure high Maintainability and {